

The diagram illustrates the sulfur cycle with the following components and reactions:

- Sulfur Species:** HS^- , $\text{S}(0)$, SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, $\text{S}_4\text{O}_6^{2-}$, SO_4^{2-} , APS .
- Enzymes and Pathways:**
 - $\text{SO}_3^{2-} \rightarrow \text{HS}^-$: DsrMKJOP (releasing e^-)
 - $\text{HS}^- \rightarrow \text{S}(0)$: FccAB/Sqr
 - $\text{S}(0) \rightarrow \text{SO}_3^{2-}$: SOR (releasing e^-)
 - $\text{SO}_3^{2-} \rightarrow \text{S}_2\text{O}_3^{2-}$: PhsABC
 - $\text{S}_2\text{O}_3^{2-} \rightarrow \text{HS}^-$: PhsABC
 - $\text{S}_2\text{O}_3^{2-} \rightarrow \text{SO}_4^{2-}$: SoxYZXAB (dashed), SoxYZXABCD (solid)
 - $\text{S}_4\text{O}_6^{2-} \rightarrow \text{SO}_4^{2-}$: TetH (dashed), SoxB (solid)
 - $\text{SO}_4^{2-} \rightarrow \text{APS}$: Sat
 - $\text{APS} \rightarrow \text{SO}_3^{2-}$: AprAB (releasing e^-)
 - $\text{SO}_3^{2-} \rightarrow \text{S}_4\text{O}_6^{2-}$: TetH (dashed), SoxB (solid)
 - $\text{S}_4\text{O}_6^{2-} \rightarrow \text{S}_2\text{O}_3^{2-}$: TsdA/DoxDA (solid), Ttr/Otr (dashed)
 - $\text{S}(0) \rightarrow \text{S}_4\text{O}_6^{2-}$: SoxB (solid), TetH (dashed)
 - $\text{S}(0) \rightarrow \text{SO}_3^{2-}$: SOR (dashed), sHdrAB1B2B3C1C2, DsrABCEFH (green), Sdo
 - $\text{S}(0) \rightarrow \text{HS}^-$: PsrABC/ShyABCD, SudAB/Npsr/Nsr (dashed)

Dissimilatory sulfate reduction pathway

The diagram illustrates the dissimilatory sulfate reduction pathway across a membrane. The pathway is divided into three regions: Cytoplasm, Membrane, and Periplasm.

- Cytoplasm:**
 - SO_4^{2-} is reduced to APS by the enzyme **Sat**.
 - APS is reduced to SO_3^{2-} by the enzyme **AprBA**.
 - SO_3^{2-} enters the membrane.
- Membrane:**
 - SO_3^{2-} is reduced to HS^- by the enzyme **DsrBA**.
 - HS^- is oxidized to $\text{S}(0)$ by the enzyme **DsrC**.
 - The oxidation of HS^- to $\text{S}(0)$ is coupled with the reduction of MQH_2 to MQ by the enzyme **DsrMKJOP**.
- Periplasm:**
 - The enzyme **QmoABC** is located in the periplasm, where it reduces MQH_2 to MQ .
 - Electrons (e^-) from this reaction are transferred to **AprBA** in the cytoplasm.

The diagram also shows the electron flow from **QmoABC** to **AprBA** via a dashed arrow labeled e^- .

The diagram illustrates the dissimilatory sulfide oxidation pathway in *Shewanella putrefaciens*, showing the flow of electrons from sulfide to various respiratory chains across the periplasm, membrane, and cytoplasm.

Periplasm:

- FccAB:** A cycle involving HS^- and HS_n^- mediated by **Sqr**, with **MQ** being reduced to **MQH₂**.
- S⁰:** A cycle involving R-S-SH and R-SH mediated by an unknown component (?).

Membrane:

- DsrMKJOP:** A cycle involving $\text{MQ} \rightarrow \text{MQH}_2$ and the transfer of electrons to **DsrC**.
- QmoABC:** A cycle involving $\text{MQ} \rightarrow \text{MQH}_2$ and the transfer of electrons to **AprBA**.

Cytoplasm:

- DsrEFH:** A cycle involving HS-SH and HS-S-SH .
- DsrC:** A cycle involving S-S(0) and S-SH .
- DsrBA/DsrL:** A cycle involving SO_3^{2-} and APS .

Electron Flow:

- Electrons from HS^- and S^0 in the periplasm are transferred to the membrane complexes **DsrMKJOP** and **QmoABC**.
- Electrons from **DsrMKJOP** are transferred to **DsrC** in the cytoplasm.
- Electrons from **DsrC** are transferred to **DsrBA/DsrL**, which then transfer them to **SO₃²⁻**.
- SO₃²⁻** is converted to **APS** by **AprBA**, and then to **SO₄²⁻** by **Sat**.